

YOUR OFE 2025 REPORT

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2025 Cropping Season

Field Kolb2 | NRate 74.5 lb N/ac | **Research question:** Will a clover cover crop supply 60 lbs of nitrogen?

Welcome to your 2025 farm report. This report summarizes the on-farm nitrogen experiment conducted in your Kolb2 field, where clover cover crop strips (treatment) were compared against no-cover-crop controls.



Figure 1. Kolb2 — treatment layout. · Cover crop: *Clover*

Take-home message: Across the nitrogen indicators we measured (NBI, chlorophyll, biomass %N, and grain %N), corn plants in the clover cover crop strips consistently showed higher nitrogen status than control plots, and yield monitor data showed a +9.7 bu/ac (+7.3%) difference in favor of the cover crop strips at harvest. That said, end-of-season cornstalk nitrate values fell below the 250 ppm deficiency threshold in both strips, suggesting that while the cover crop improved early-season nitrogen availability, the full 60 lb N/ac target was likely not met, making direct clover biomass sampling in 2026 the key next step.

Results Summary

Metric	Control	Cover (Treatment)	Difference	p-value
 Corn Yield(bu/ac)	133 bu/ac	142.7 bu/ac	+9.7 bu/ac (+7.3%)	spatial data†
 NBI	73.6	89.4	+15.8 (21.5%)	p < 0.001
 Chlorophyll Index	41.6	48.4	+6.8 (16.3%)	p < 0.001
 Biomass %N	2.392%	2.828%	+18.2%	n too small*
 Grain %N	1.085%	1.262%	+16.3%	p = 0.00729
 CSNT	74 ppm	57 ppm	-17 ppm	n = 1

* Biomass C:N sample sizes are small (n = 5 Control, n = 5 Treatment). Effect sizes (Cohen's d) are reported instead of p-values for that section. † Yield monitor data contains 24,539 GPS points with spatial autocorrelation — a simple p-value is not appropriate. The difference is reported as a descriptive field-level observation.

Corn Yield

The map below shows the **corn yield spatial distribution** across the Kolb2 field, displayed in bushels per acre. The red outline marks the clover cover crop (treatment) area and the blue outline marks the control strip.

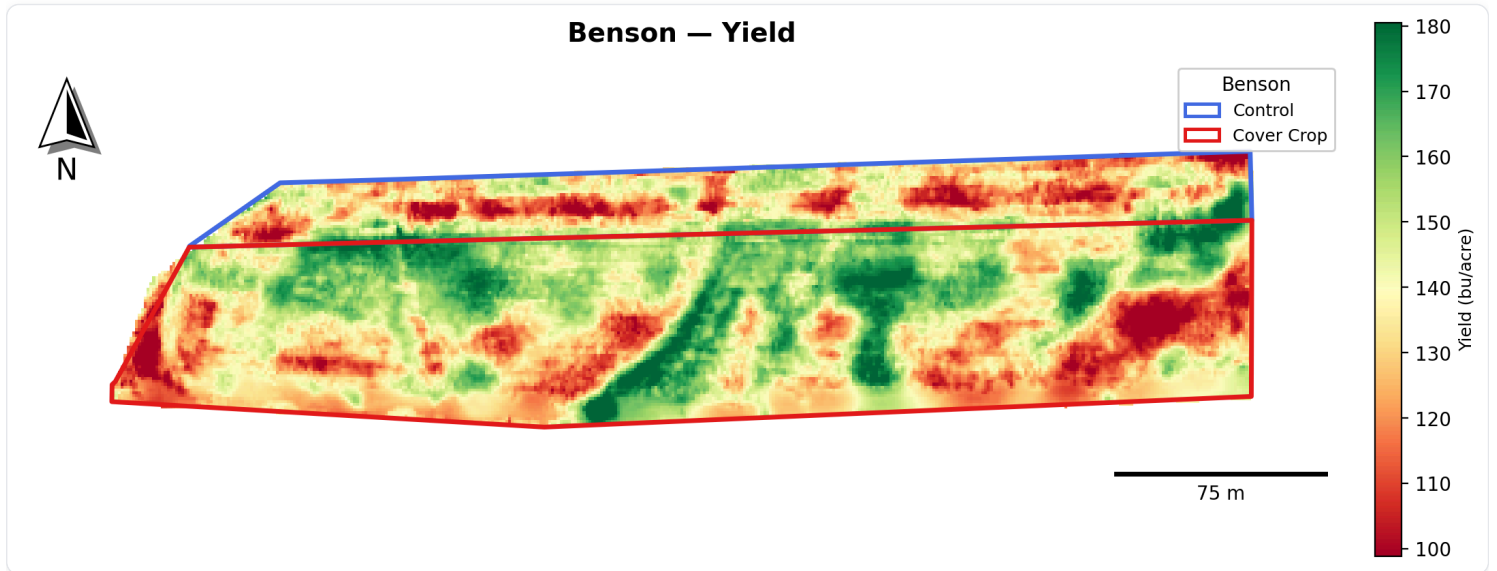


Figure 2. Corn yield map.

[Open Interactive Map](#)

TREATMENT — MEAN YIELD

142.7 ↑ +9.7 bu/ac vs Control (7.3%)

Control mean: 133 bu/ac | n = 18,417 yield monitor points

DIFFERENCE

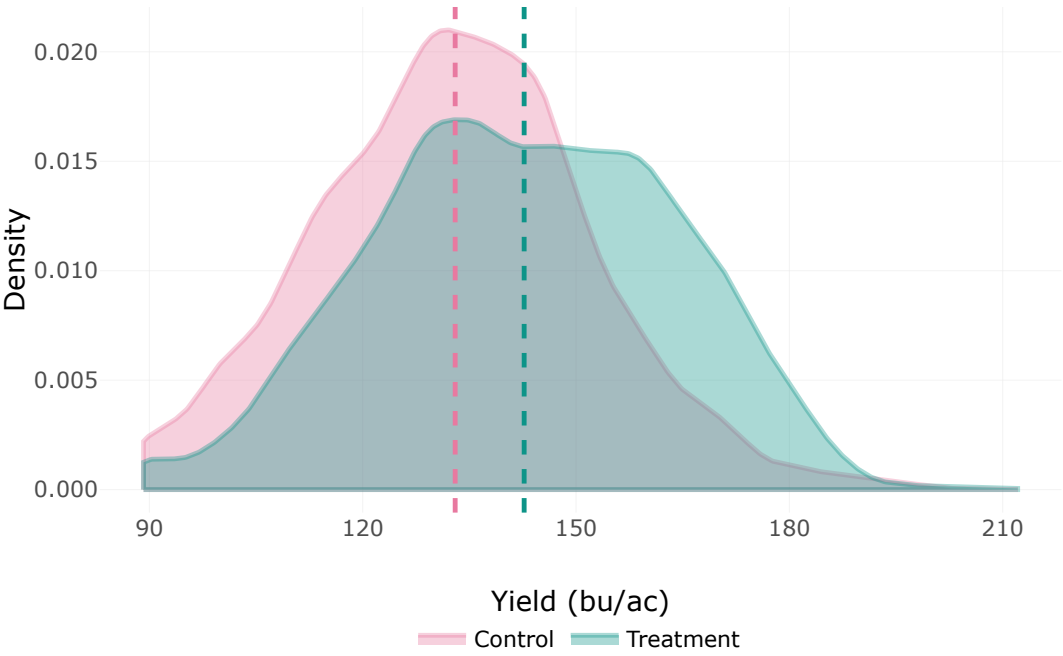
+7.3%

Yield monitor data — spatial autocorrelation expected.
Difference is descriptive, not a formal statistical test.

Density

Boxplot

Corn Yield Distribution



Nitrogen Balance Index (NBI)

The **Nitrogen Balance Index (NBI)** gives us a real-time read on how well-supplied with nitrogen each corn plant is during the growing season. It works by measuring both chlorophyll (a sign of nitrogen sufficiency) and flavonoids (stress compounds that build up when the plant is nitrogen-stressed) in the same leaf. A higher NBI means the plant has plenty of nitrogen and is using it efficiently. We took two leaf readings per plant with the Dualex sensor, averaged them, and compared the clover strips against the control using a statistical test to see if the difference was real.

TREATMENT

89.4 ↑ +15.8 vs control (21.5%)

Control mean: 73.6

EFFECT SIZE

21.5%

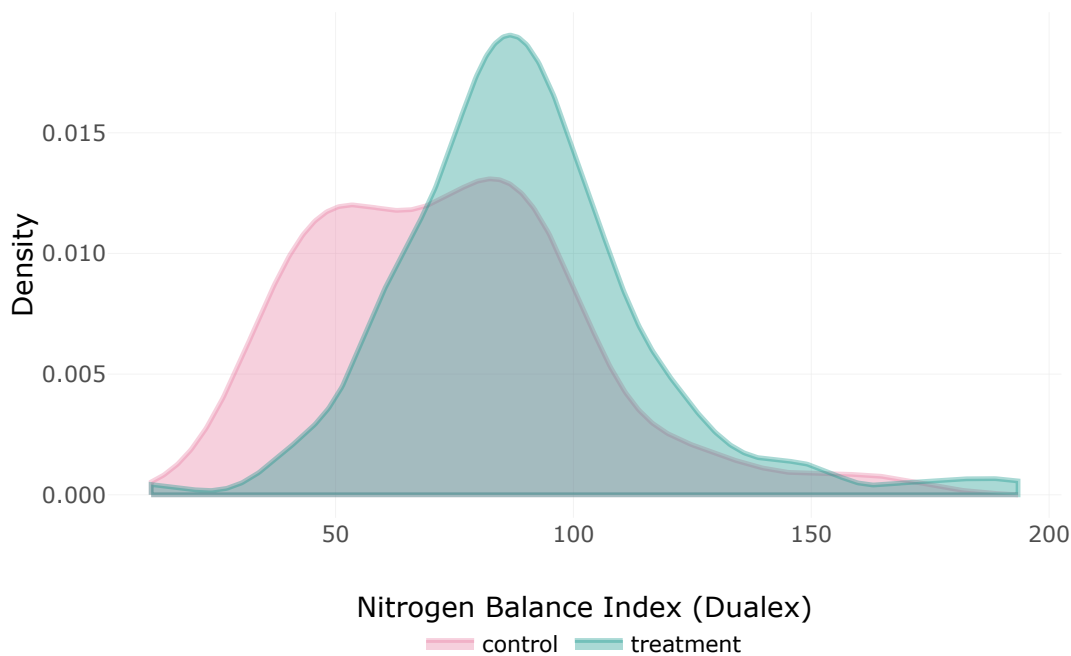
relative difference treatment vs control

$p < 0.001$ ✓

Density

Boxplot

Nitrogen Balance Index (NBI)



Chlorophyll (Chl)

Leaf greenness is one of the most direct, visible signals that a corn plant has enough nitrogen. The Chlorophyll Index, measured with the same Dualex sensor as the NBI, captures this at the leaf level. A higher value means a greener, more photosynthetically active leaf, the kind of leaf that comes from a plant with good nitrogen access. We averaged two readings per plant and compared clover vs. control strips with a statistical test (t-test). When both the NBI and the chlorophyll are elevated in the same strips, that's a consistent and reliable signal.

TREATMENT

48.4 ↑ +6.8 vs control (16.3%)

Control mean: 41.6

EFFECT SIZE

16.3%

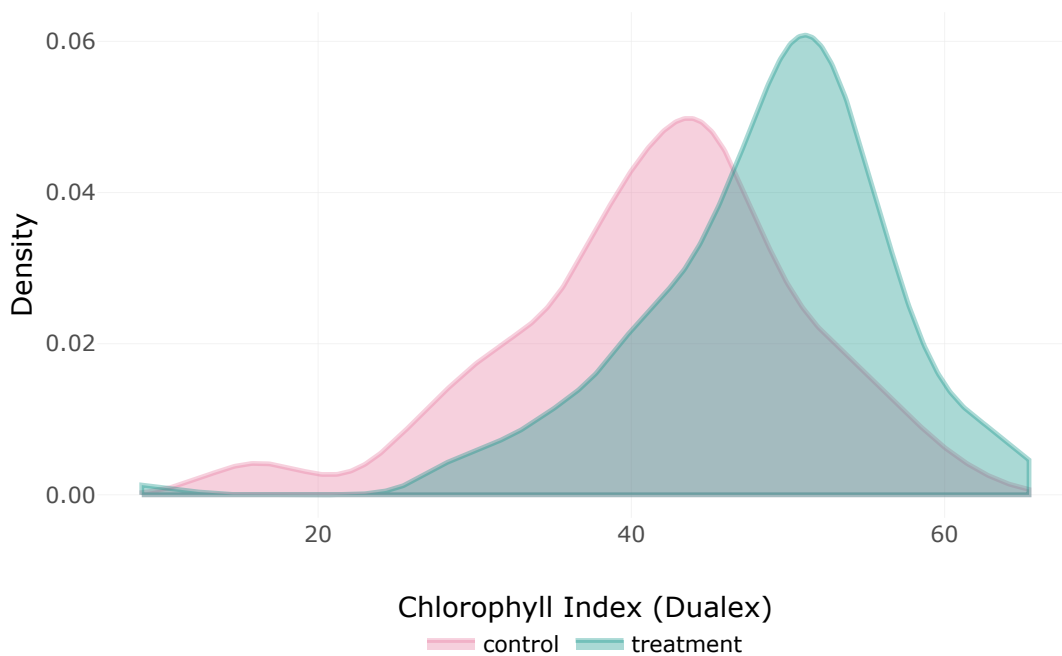
relative difference treatment vs control

p < 0.001 ✓

Density

Boxplot

Chlorophyll (Chl)





C:N in Corn Biomass

Corn biomass was collected and sent for laboratory C:N analysis. This tells us how much nitrogen the plant had absorbed and incorporated into its tissue up to that point in the season, a more direct measure than the leaf sensor readings. We only collected a few biomass samples (5 control, 5 treatment), which is too few for a reliable statistical test. Instead, we report an “effect size” (Cohen’s d). It answers a simple question: is the gap between the cover crop and control strips bigger than the normal differences you see from one sample to the next within the same strip? A value near 0 means the two strips look about the same. A value above 0.8 means the gap clearly stands out from that normal variation.

Total Nitrogen in Biomass (%N)

A higher %N in the tissue means the plant had better access to nitrogen during the early-to-mid season growth window. The clover strips came in notably higher than the control, consistent with the in-season Dualex readings.

TREATMENT — MEAN %N (BIOMASS)

2.828 ↑ +18.2% vs Control

Control mean: 2.392

COHEN'S D (EFFECT SIZE)

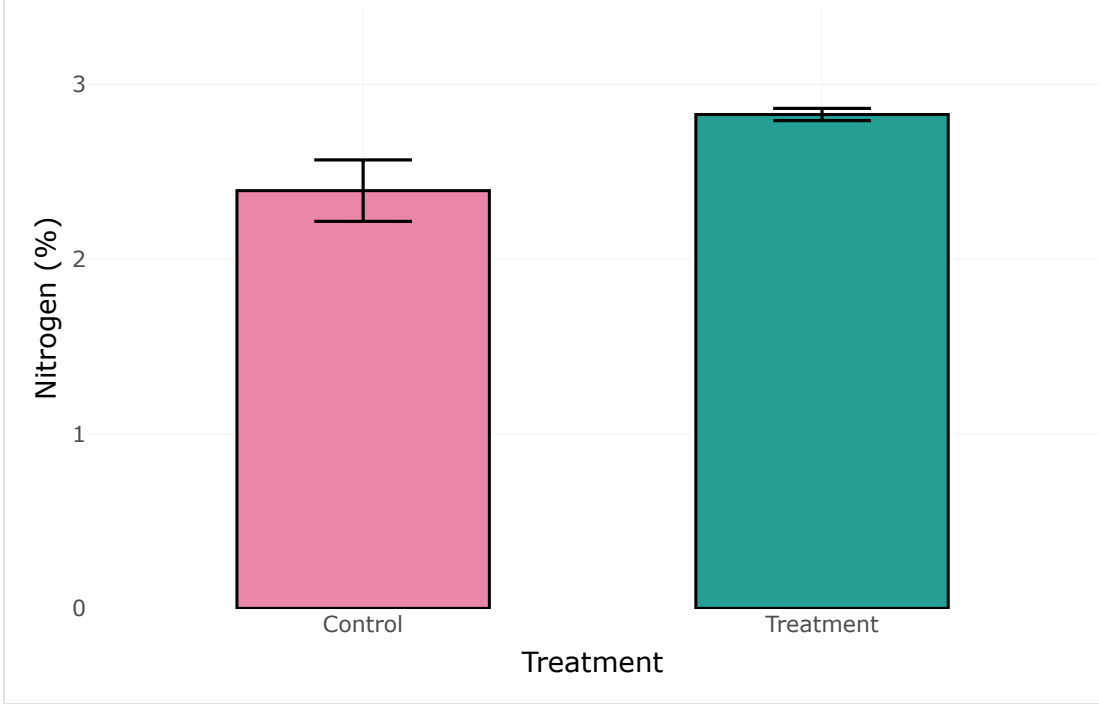
1.53

n = 5 Control · n = 5 Treatment

Bar Chart (%N)

Raw Data (%N)

N in Corn Biomass (Mean %N ± SE)



C:N Ratio — Biomass

TREATMENT — MEAN C:N RATIO

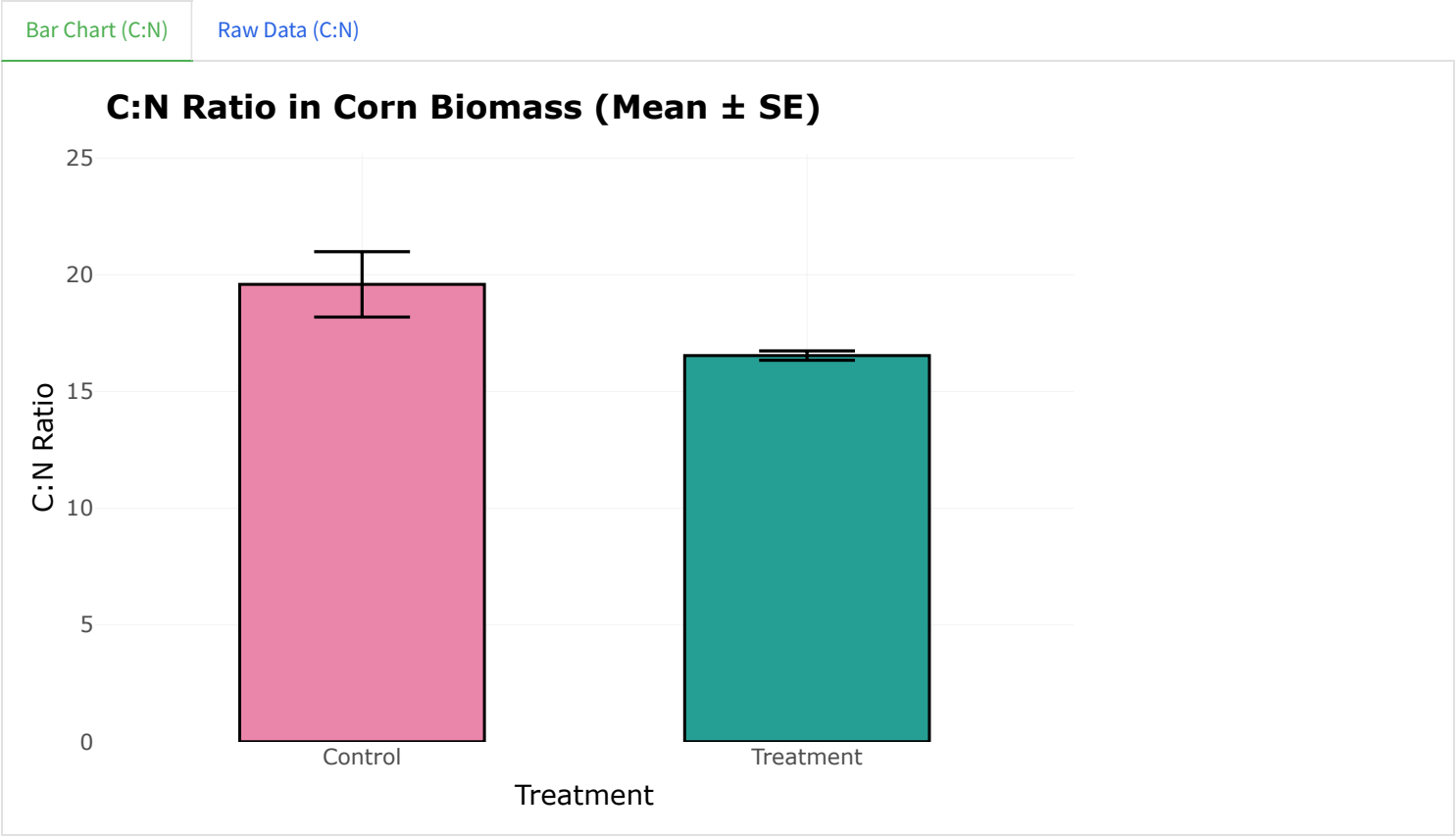
16.54 ↓ -15.6% vs Control

Control mean: 19.59

COHEN'S D (EFFECT SIZE)

1.36

Lower C:N = more nitrogen-rich tissue



C:N in Corn Grain

Grain samples were collected and analyzed for C:N content. Grain %N is closely related to grain protein. The C:N ratio expresses how much carbon there is for each unit of nitrogen, so a lower ratio means more nitrogen-rich grain. If the clover strips show higher %N and a lower C:N, it suggests those plants had more nitrogen available during grain fill.

Total Nitrogen in Grain (%N)

TREATMENT

1.262 ↑ +0.177 vs control (16.3%)

Control mean: 1.085

EFFECT SIZE

16.3%

relative difference treatment vs control

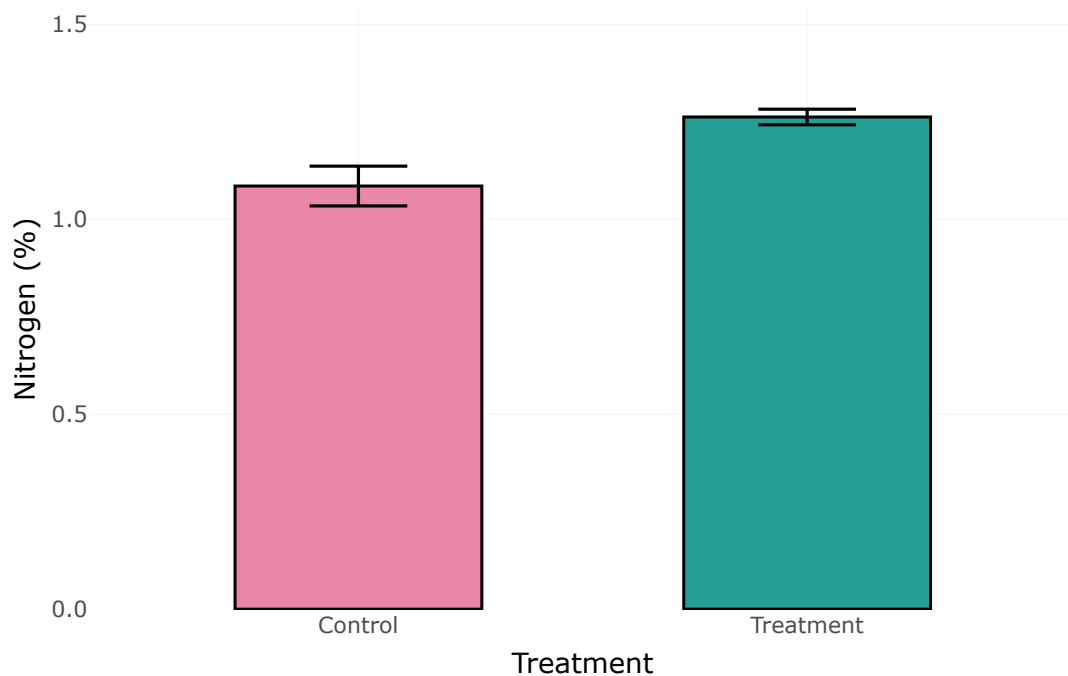
p = 0.00729 ✓

Bar Chart (%N — Grain)

Boxplot (%N — Grain)

Raw Data (%N — Grain)

N in Corn Grain (Mean %N ± SE)



C:N Ratio — Grain

TREATMENT

37.02 ↓ -6.32 vs control (-14.6%)

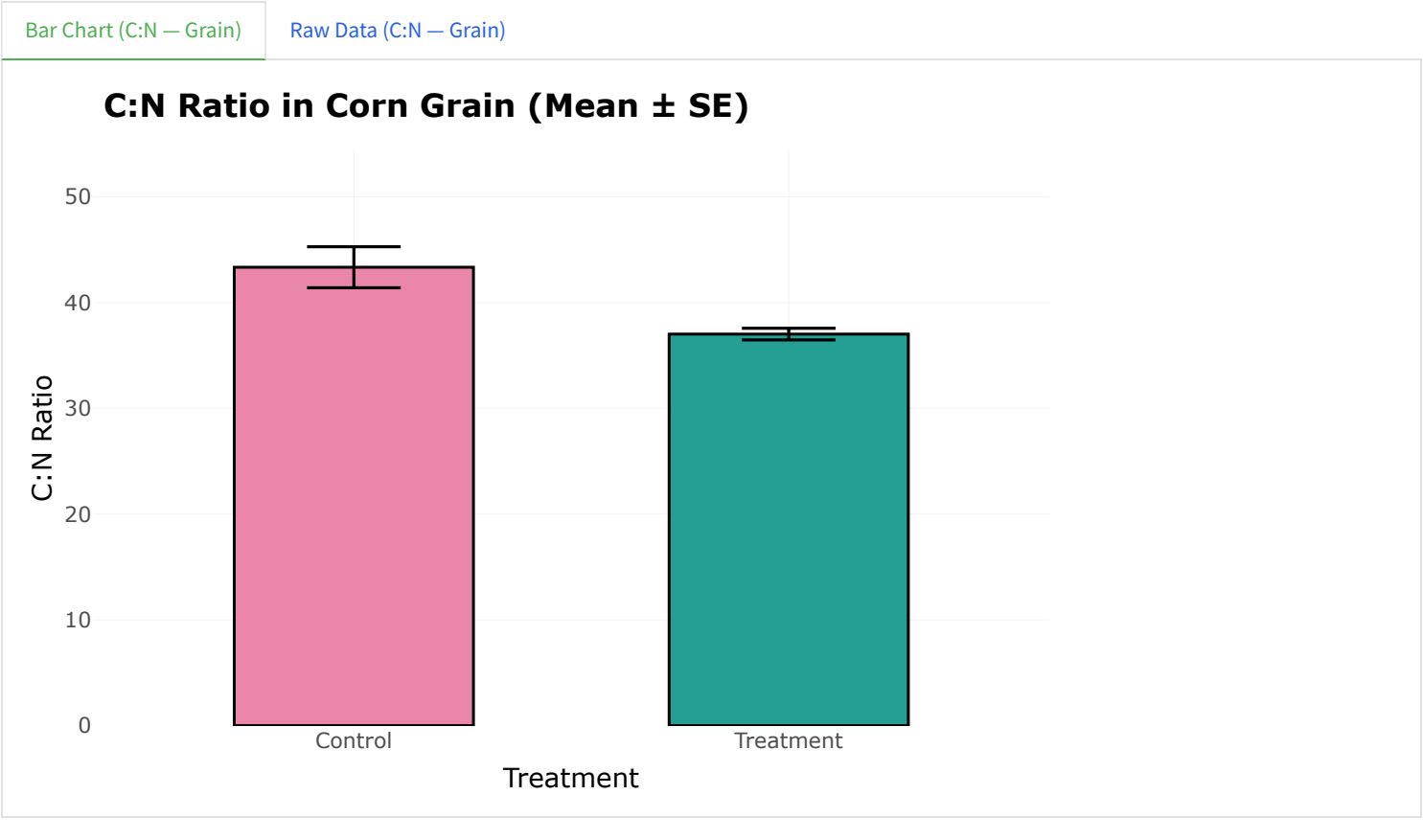
Control mean: 43.34

EFFECT SIZE

-14.6%

relative difference treatment vs control

p = 0.0101 ✓





Cornstalk Nitrate Test (CSNT)

The Cornstalk Nitrate Test (CSNT) measures nitrate concentration in the lower cornstalk at the end of the season. It is a post-harvest diagnostic tool: values **below 250 ppm indicate insufficient nitrogen**, while the **optimal range is between 750 and 2000 ppm**. Because only **one sample per treatment** was collected, no statistical comparison is possible — this is a field-level observation.

CONTROL

74 ppm

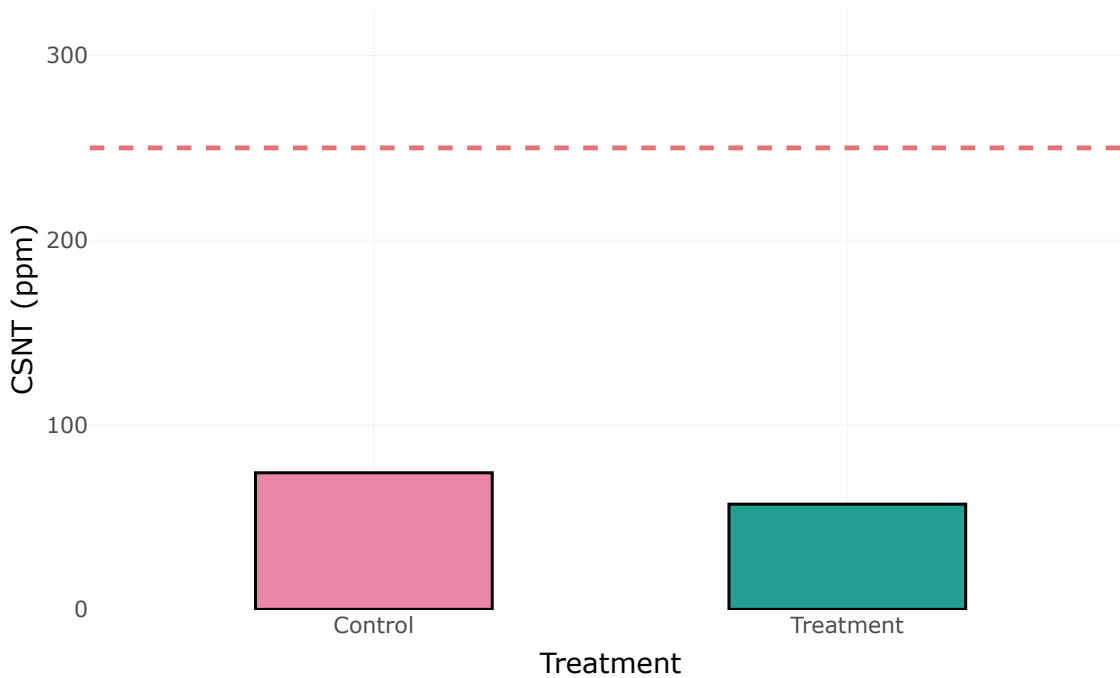
⚠️ Below 250 ppm deficiency threshold

TREATMENT

57 ppm

⚠️ Below 250 ppm deficiency threshold

Cornstalk Nitrate Test (CSNT) — Benson Kolb2



 Soil Health Assessment

Soil health was assessed using the **Cornell Soil Health Test**. The overall quality score was **78/100 (High)**. Key items to watch: **Soil pH** scored 28 (Low, orange) at 5.7, below the optimal 6.2 - 6.8 range for corn, which can reduce nutrient availability and should be addressed with lime. **Soil Respiration** (score 56) and **Active Carbon** (score 54) are both in the Medium range, indicating microbial activity and labile carbon could improve. Physical and most biological indicators are strong: Aggregate Stability (100), Organic Matter (100), and AWC (93) reflect the benefits of no-till management. Aluminum (105.1 ppm) is flagged as excessive, which is linked to the low pH.

Comprehensive Assessment of Soil Health

From the Cornell Soil Health Laboratory, Department of Soil and Crop Sciences
School of Integrative Plant Science, Cornell University, Ithaca, NY 14853
<https://soilhealthlab.cals.cornell.edu>



Agricultural Service Provider:
Alex Yole
ay327@cornell.edu

Sample ID: CC34
Field ID: Benson Kolb
Date Sampled: 06/11/2025
Tillage: no till
Coordinates: Latitude: 42.635928
Longitude: -76.428336

Measured Soil Textural Class: **silt loam**
Sand: **31%** - Silt: **55%** - Clay: **14%**

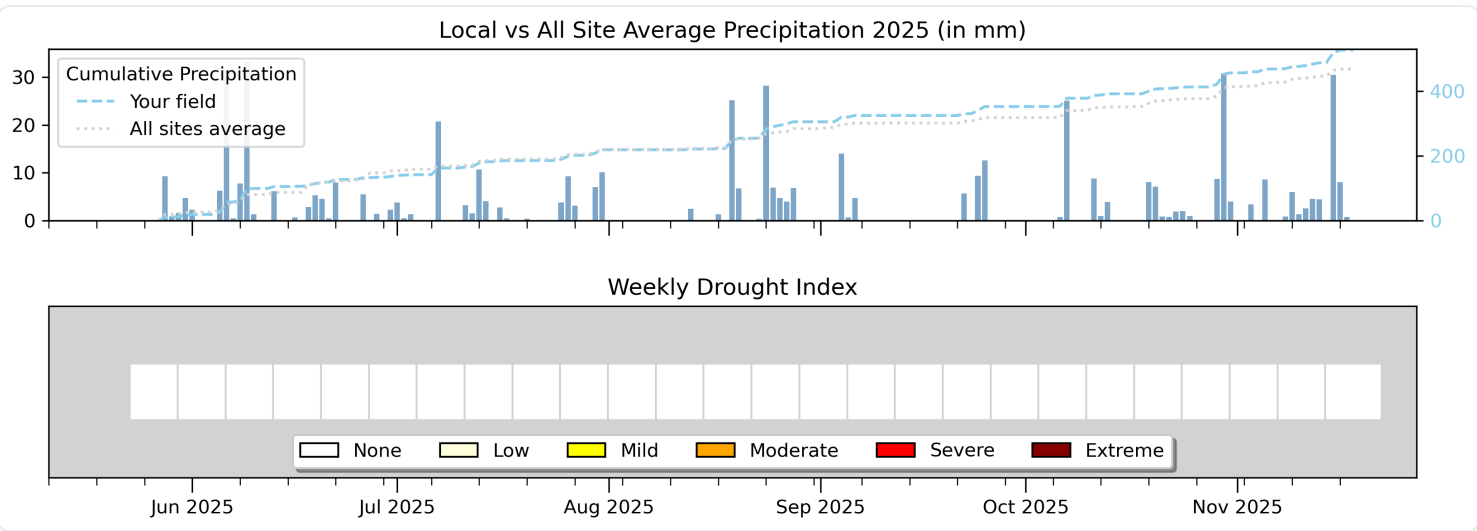
Group	Indicator	Value	Rating	Constraints
physical	Predicted Available Water Capacity	0.25	93	
physical	Surface Hardness	112	73	
physical	Subsurface Hardness	267	62	
physical	Aggregate Stability	90.8	100	
biological	Organic Matter Soil Organic Carbon: 3.36 / Total Carbon: 3.4 / Total Nitrogen: 0.32	5.5	100	
biological	ACE Soil Protein Index	9.6	83	
biological	Soil Respiration	0.6	56	
biological	Active Carbon	518	54	
chemical	Soil pH	5.7	28	
chemical	Extractable Phosphorus	5.1	100	
chemical	Extractable Potassium	100.9	100	
chemical	Additional Nutrients Ca: 822.7 / Mg: 167.1 / S: 19.8 Al: 105.1 / B: 0.09 / Cu: 0.12 Fe: 9.8 / Mn: 3.2 / Zn: 0.4		88	

Overall Quality Score: **78 / High**

 **Open Full Soil Health Report (PDF)**

Weather & Drought Conditions

Cumulative precipitation at your field compared to the all-site average, alongside the weekly drought index, from late May through mid-November 2025. Your field received about 530 mm (about 21 inches) of rain over this period, somewhat more than the all-site average of about 470 mm. The two tracked closely through mid-August; from late August onward your field received more rain than the average site. Rain was not evenly spread: there were several large events of more than 25 mm (1 inch) in a day, in early June, late August, October and November, and a mostly dry stretch in the first half of August. Even so, the weekly drought index stayed at None for every week shown, so no drought stress was recorded.





Conclusions

LEAF NITROGEN STATUS (NBI)

Cover crop strips showed a **21.5% higher NBI** compared to Control (89.4 vs 73.6). Chlorophyll was also higher by **16.3%** (48.4 vs 41.6). Both in-season readings point in the same direction: corn plants in cover crop strips had consistently better nitrogen status during the growing season.

GRAIN NITROGEN (%N)

At harvest, grain %N was **+16.3% higher** in cover strips (1.262% vs 1.085%; $p = 0.00729$; $n = 10$ control, 20 treatment). This reflects the full-season nitrogen story: cover crop plants not only showed better in-season status but also accumulated more nitrogen in the grain itself.

CORN YIELD — COVER VS NO COVER

Yield monitor data showed cover crop strips averaged **142.7 bu/ac** vs **133 bu/ac** in control strips, a difference of **+9.7 bu/ac (+7.3%)**. Given the spatial autocorrelation inherent in yield monitor data, this is a descriptive field observation rather than a formal statistical result, but the direction is consistent with all in-season nitrogen indicators.

BIOMASS NITROGEN (%N)

Laboratory analysis of corn biomass showed cover strips had **+18.2% more nitrogen** in their tissue (2.828% vs 2.392%). Cohen's $d = 1.53$, indicating a large effect. With $n = 5$ control and $n = 5$ treatment samples, formal statistical testing is not appropriate, but the magnitude of the difference is notable and consistent with the Dualex readings.

CSNT — N LIMITING IN BOTH STRIPS

End-of-season stalk nitrate was below the 250 ppm deficiency threshold in both treatments (Control: **74 ppm**; Cover: **57 ppm**). This indicates the corn crop was nitrogen-limited regardless of treatment. The 60 lb/ac clover N target was likely not met this season; direct clover biomass sampling would help quantify actual N supply in 2026.